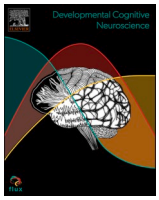
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A systematic review of digital and wearable sensor data in the ABCD study: Insights, methods, and emerging opportunities

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ABSTRACT

Research into key adolescent health domains, like problematic screen-based media use, sleep health, and physical activity, are frequently limited by error-prone self-report-based measures. To improve the measurement of such behaviors, the Adolescent Brain Cognitive Development Study (ABCD), a diverse longitudinal cohort study of nearly 12,000 U.S. youth (ages 9–10 at Baseline), includes remote sensing protocols during data collection. Consenting participants provided up to three weeks of Fitbit actigraphy data, beginning at the year-two follow up, and smartphone usage data, beginning at the year-four follow up, collected via the Effortless Assessment of Research Systems (EARS) smartphone application. This systematic review aimed to identify and synthesize all published empirical studies which have utilized the ABCD Study's passive sensor data, identifying 45 articles investigating measurement validity, and associations with adolescent health, development, and neurocognitive functioning. Fitbit devices broadly reproduced estimates of sleep and physical activity obtained from research-grade sensors. Smartphone sensor data were internally consistent, reproduced expected daily usage patterns, and were moderately correlated with self-report. Most substantive studies utilized Fitbit data and reported cross-sectional associations between sensor-derived single-item summary statistics and survey-based measures of mental and behavioral health, neurocognitive functioning, or environmental risk factors. A smaller number of studies utilized longitudinal designs, within-person analyses, machine learning approaches, and/or genetic or neuroimaging methods to reduce confounding and enhance biological and etiological insight into these observed relationships. While published findings were largely incremental in nature, ABCD's passive sensing resources nonetheless represent a powerful, widely available resource for incorporating sensor-based measures into adolescent health research.

1. Introduction

Over the past ten years, concerning behavioral health trends have emerged among adolescents. Though they are less likely to engage in risk behaviors like alcohol, drug use, or risky sexual behavior, today's adolescents exhibit far higher rates of mood and anxiety disorders than

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prior cohorts at the same age (Miech et al., 2025; Youth Risk Behavior Survey, 2024). Longstanding behavioral health challenges faced by adolescents, including inadequate sleep and excessive sedentary behavior, also appear more prevalent in contemporary adolescents. As of 2023, fewer than half of U.S. high school students reported regularly meeting recommended physical activity levels and fewer than one quarter reported regularly sleeping 8 or more hours per night (Youth Risk Behavior Survey, 2024). While causes are likely multifactorial, they may in part reflect mass smartphone adoption by teenagers in the early 2010s. Widespread smartphone adoption closely coincides with rising rates of adolescent anxiety and depression and significant associations have been observed between time spent using digital devices and youth mental and behavioral health concerns (Carter et al., 2016; Haghjoo et al., 2022; Santos et al., 2023). The troubling trends have been collectively deemed a "youth mental health crisis" by leading public health agencies such as the CDC, the U.S. Surgeon General's office, and the World Health organization, highlighting the growing alarm over youth mental health from concerned parents, clinicians, policy makers, and adolescents themselves (CDC, 2024; World Health Organization, 2025; U.S. Surgeon General's Advisory, 2024).

Research into the important areas of physical activity, sleep, and screen media activity (SMA) face substantial challenges. Studies to date have relied almost entirely on simple self-report-based measures of SMA, sleep, and physical activity. These measures often omit important nuances, like the timing and context of activity, and are subject to retrospective reporting and recall bias, which may introduce error or bias into usage measures (Paulus and Vazire, 2007). Indeed, studies comparing self-reported device use to sensor-based estimates find adolescents typically underreport SMA use by multiple hours per day (Alexander et al., 2024). Such measurement challenges are not limited to SMA exposure. Other behavioral measures, such as sleep and physical activity, frequently show large discrepancies between self-reports and gold-standard sensor-based measures (Erwin and Bashore, 2017; Healey et al., 2020; Lauderdale et al., 2008; Prince et al., 2008).

Sensor-based data from digital and wearable devices offer significant advantages over traditional self-report measures of SMA, physical activity, and many of their behavioral correlates. Such data are passively recorded in real time, rendering them robust to reporting and recall biases and offering improved validity over retrospective report measures (Harari et al., 2016). Device-based sensors further facilitate continuous, high dimensional data collection without overburdening participants, improving acceptability over survey-based ecological momentary assessment methods while enabling rich, temporally granular measures of participant behaviors, such as the timing and context of smartphone usage, physical activity, or sleep (Alexander et al., 2024; Godino et al., 2020; Harari et al., 2016).

Despite these advantages, the relative novelty of the approach, the high cost of the technology, and the logistical and technical complexity of additional study management and data processing requirements represent significant barriers to implementation. Consequently, sensor-based studies have primarily been conducted in small and often unrepresentative samples of adolescents. The Adolescent Brain Cognitive Development Study (ABCD) is an ongoing study of adolescent neurodevelopment with a diverse sample which approximates the demography of the United States in nearly 12,000 adolescents. Smartphone and wearable sensors have been employed in the extensive research protocol of this study. In combination with its many other protocol elements, the sensor-based data in the ABCD Study provides exceptional opportunities to advance our understanding of the role of adolescent digital technology use and physical activity in adolescent mental and behavioral health trends.

Here we review ABCD studies that were published through July 2025 that utilized passive sensing to provide a more nuanced understanding of behavioral and physical health trends in adolescence. While others recently reviewed self-reported SMA behaviors in ABCD (Nagata, Lee, et al., 2025), we focus on sensor-based data given the potential improvements in data quality, quantity, and analytical methods applied. In doing so, we aim to identify contributions of ABCD passive sensing thus far, while also looking ahead to promising future directions for use of these data. This review spans the breadth of ABCD, including methods of passive sensing (measurement validity, machine learning), major domains assessed within ABCD (mental health, physical health, screen media activity, neuroimaging, cognition), and considerations for health disparities. Highlighted findings can provoke future research questions for this dataset concurrently with clarifying modern adolescent digital and behavioral health trends and risk factors to inform policy and public health.

Terminology

- ABCD Study = Adolescent Brain Cognitive Development Study, a 10 + year longitudinal study of nearly 12,000 youth followed annually starting at ages 9–10
- EARS = Effortless Assessment of Research Systems, a research-based app which runs continuously in the background on a smartphone to log smartphone activity
- Fitbit = A commercial wearable fitness tracker that utilizes passive sensing technology and proprietary algorithms to estimate physical, sedentary, and sleep behaviors
- Sensor-based = mobile and wearable technologies that rely on sensors to track behaviors (e.g., EARS; Fitbit); also referred to as passive sensing

2. Methods

The initial review of articles was conducted according to the Preferred Reporting Items for Systematic Review and Meta Analyses, PRISMA, guidelines (Page et al., 2021). Search parameters and keywords were specified using the SPIDER framework for systematic formulation (Cooke et al., 2012). The sample of interest was identified as ABCD study participants. The phenomena of interest were findings related to the validity and acceptability of sensor and wearable-based measures and their associations with adolescent mental and behavioral health outcomes. The research design, evaluation, and research methods of interest were studies using associational, quantitative designs to investigate Fitbit or smartphone-based sensor measures in the ABCD study.

A search was conducted on 06/30/2025 on online databases Medline, Psycinfo, and Web of Science using the standardized keywords included in **Appendix A**. An additional search was conducted on 07/21/

2025 using a database of ABCD publications filtered to the "novel technologies" category (https://abcd-study.shinyapps.io/abcd-publications/).

An initial review of article titles removed duplicate articles, non-human subjects research, review papers, meta-analyses, and research conducted entirely using participants residing outside of the United States (which necessarily did not include ABCD subjects). Article titles, abstracts, and methods sections were then evaluated for inclusion in a six-step screening process:

1. Articles published in 2021 or later were included, as this was the first year in which ABCD wearables data were made available for research.
2. Articles needed to be available in English to ensure thorough review by the primarily English-speaking research team.
3. Original, empirical articles published in peer-reviewed journals were included; reviews, meta-analyses, preprints, and unpublished theses were not included in the review.
4. While studies utilizing multiple samples were included, at least 25% of the analytic sample was required to be ABCD study participants.
5. Studies were required to utilize ABCD's remote sensing measures, either passive smartphone or Fitbit data.
6. Quantitative or mixed-methods studies were included; fully qualitative studies were not retained in the review.

Articles meeting all review criteria were further coded for sensor data source (i.e., smartphone, Fitbit, or both), sensor measure (e.g., sleep, physical activity, smartphone use), primary outcome(s) (e.g., health disparities, mental health), and analytic approach (e.g., machine learning). Articles were then sorted into eight content areas: measurement validity, machine learning prediction, physical health, mental health, digital media usage, neuroimaging, cognition, and health disparities. Content groupings were not mutually exclusive, with articles including content across multiple domains (e.g., articles discussing both the validity of sensor measures and substantive mental or behavioral health findings) included in up to three different groups. After grouping, articles were read in full by content experts identified from among the ABCD Novel Technologies Workgroup, who drafted article summaries and qualitative syntheses of each content domain. Methods employed here were not pre-registered.

3. ABCD sensor-based measures

The ABCD Study is a 10 + year longitudinal dataset of 11,878 diverse U.S. youth. Annual assessments cover mental health, physical health, substance use, social relationships, environmental factors, and more. Biannually, youth additionally complete a 60-minute neurocognition battery and 90-minute multimodal neuroimaging session. Participants and a parent report on screen use (by weekday, weekend; mode of use such as streaming, single player video games, etc.), screen behaviors, and problematic use of their mobile device, social media use, and gaming. They also detail their physical activity and sleep behaviors (Garavan et al., 2018; Karcher and Barch, 2021).

Beginning at the 2-year follow-up assessment (~age 12), participants are invited to wear a Fitbit Charge Series device for three weeks to monitor all aspects of physical activity (including sleep). Specific device models varied by assessment wave, with participants provided the Fitbit Charge HR at year 2, the Fitbit Charge 2 at year 4, and Charge 4–6 model devices at year 6. Wave 4 Fitbit data collection is offered every other year to coincide with neuroimaging sessions. This data has provided high-resolution indicators of how sleep and physical activity behaviors relate to neurodevelopment and health outcomes across adolescence. Recommendations for data use are available in Wing et al. (2022).

Starting at the 4-year follow-up (~age 14), youth are invited to download a passive sensing app (Effortless Assessment of Research Systems, EARS) onto their personal smartphone for three weeks to objectively log SMA, including specific app use, timing, and duration (Lind et al., 2023). Smartphone sensor data is collected annually. Smartphone sensor data were collected concurrently with Fitbit data whenever possible (Lind et al., 2023). The EARS app does not log content or specific information regarding what adolescents see, but logs start/stop times and the name of the foregrounded application for each instance of smartphone use. Keystroke counts, typing duration, and start and stop times of keyboard use instances are also recorded as a measure of more active device engagement.

Data collection varied between Apple iOS and Android devices due to platform differences, particularly iOS-specific limitations to device data collection by third-party applications like the EARS App. Most notably, only keyboard-based measures were initially available from iOS users, while direct measures of application usage time were limited to Android devices. Additionally, while on Android devices keyboard measures could be collected from the device's standard keyboard, iOS users were required to use an EARS-specific keyboard overlay to facilitate collection of keyboard measures (Alexander et al., 2024).

For both Fitbit and EARS data, the ABCD Data Releases include raw (exact logged data provided by the devices) and summary data (e.g., average moderate-to-vigorous physical activity per day; average smartphone app usage per day).

4. Study quality assessment

All included studies were individually rated for quality of evidence and risk of bias using the Quality Assessment Tool for Observational Cohort and Cross-Sectional Studies (L.-L. Ma et al., 2020). This tool was chosen because of its flexibility between cohort and cross-sectional designs. Notably, as all studies come from the ABCD dataset, to a certain extent all studies had similar strengths (large cohort, objective passive sensing metrics) and similar weaknesses (sampling biases, some heterogeneity in data collection across study sites); quality assessments were conducted with this in mind, with ratings assessed by paper-specific methodological choices rather than to shared elements of the overarching ABCD design. If papers included both passive sensing and self-report models, quality assessment focused only on passive sensing models to adhere to the scope of the review.

Individual scales are included in the **Supplement**. Assessments were conducted by JDA, NEW, and BD. Inter-rater reliability was assessed by double-scoring 10 randomly selected studies. If raters disagreed on overall quality assessment ratings, they would discuss the individual merits and come to a consensus agreement, prior to completing the remaining assessments. Raters were not allowed to be authors on an assessed paper.

5. Results

A PRISMA flow diagram reporting the number of records identified, screened, excluded, and retained in the review is provided as Fig. 1**.** The initial article search of Medline, PsycINFO, Web of Science, and the ABCD publication database yielded 708 candidate articles. After screening article titles to remove duplicate articles, non-human subjects research, reviews and meta-analyses, and research conducted entirely using participants from outside the United States, 361 articles were retained for further screening. Articles were screened by three independent coders. A subset of 90 articles were independently reviewed by a minimum of two coders; inter-rater reliability for ultimate article inclusion/exclusion was 100% between each pair of raters.

A total of 47 articles initially met inclusion criteria, though two were excluded during full-text reading (one preprint initially misidentified as having been published in a peer-reviewed journal and one duplicate article indexed under different titles on Medline and PsycINFO) leaving **45 articles for inclusion in the review**. During content coding, eight articles were identified as focused partially or primarily on measure validation, nine on associations with neuroimaging outcomes, eight on

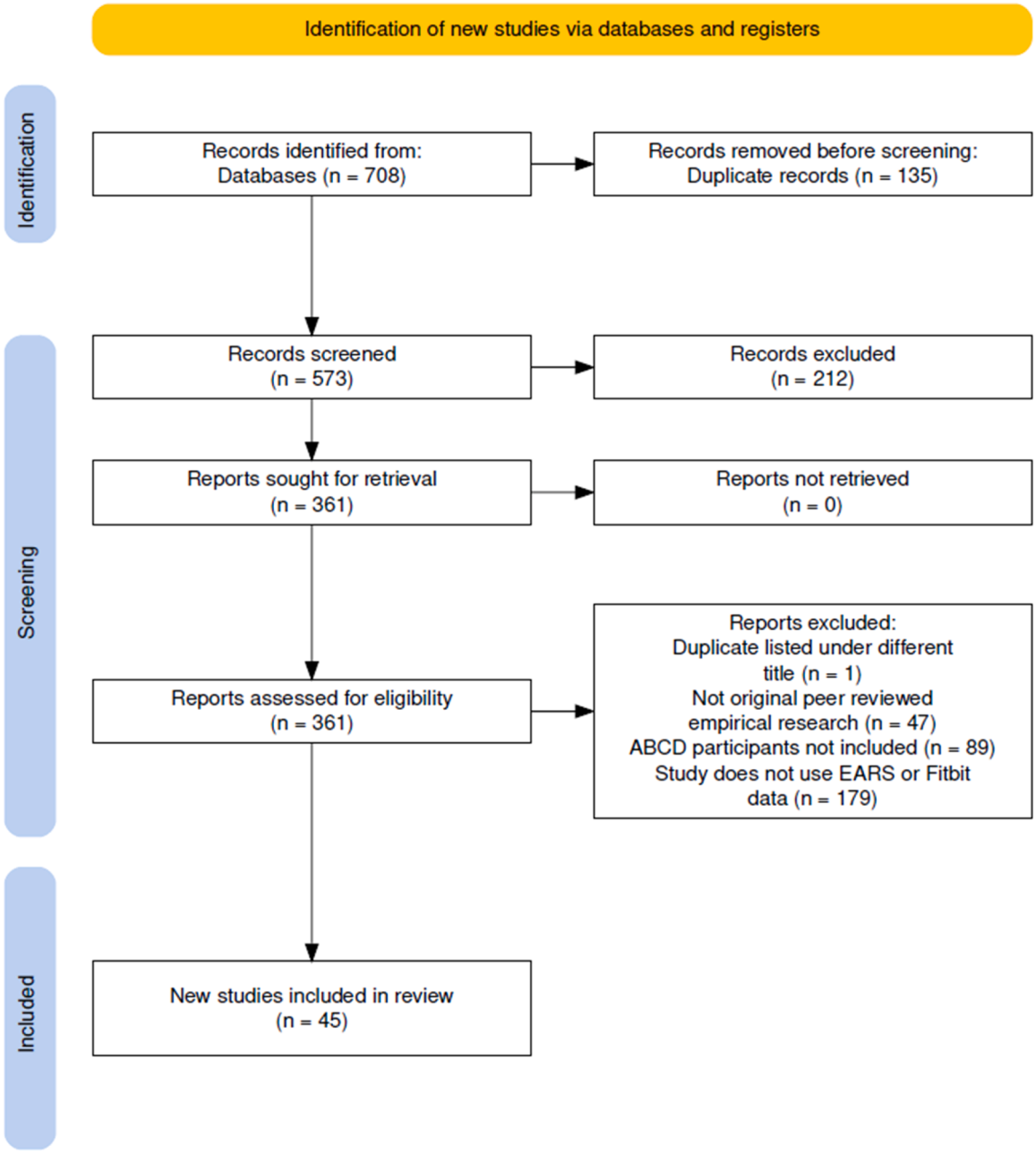


Fig. 1. Flow diagram describing the article screening and selection process via Preferred Reporting Items for Systematic reviews and Meta-Analyses (PRISMA) diagram. This diagram was produced using PRISMA2020, a Shiny app for producing PRISMA compliant flow diagrams (Haddaway et al., 2022).

associations with physical health, 15 on associations with mental health, four on the development of machine learning prediction algorithms, nine on health disparities, three on associations with cognitive outcomes, and seven on associations with screen-based media usage. A review of each content area is provided below. A list of articles by content domain are provided in Table 1.

6. Sensor-based measurement validity

6.1. Fitbit

Prior to ABCD Fitbit data collection, an initial pilot study was conducted in a sample of 59 early adolescents was completed by ABCD investigators to parallel the ABCD cohort (Godino et al., 2020). This study compared sleep and physical activity measures obtained concurrently from a Fitbit HR device and research-grade sensors, including an electrocardiogram, calorimeter, portable polysomnography system, and video recorded step counts, during a lab-based physical activity battery and home-based sleep assessment. On average, compared to gold standard measures, Fitbit HR reads underestimated step counts by 9.9%, underestimated heart rates by 3.3%, overestimated energy expenditure by 11.4%, underestimated METs by 11.3%, underestimated total sleep time by 14 min, and overestimated nighttime awakening by 9 min. Accuracy in classifying physical activity as sedentary, light, moderate, or vigorous was 84.8% while accuracy in distinguishing sleep from nighttime waking was 92.1%. Notably, standard errors were not reported, though the study's small sample size suggests there may be considerable uncertainty in these point estimates.

A subsequent pilot study of 154 ABCD participants explored acceptability and participant compliance with Fitbit data collection and developed a data pre-processing pipeline to identify and exclude invalid data (Wing et al., 2022). The vast majority of participants reported

Table 1
Reviewed Articles, Content Grouping, and Primary Sensor-Related Findings.

Article Title/Citation	Sensor; Measure	Study Outcome	Grouping	Sample Size	Primary Sensor-Based Findings
Individual-level experiences of structural inequity and their association with subjective and objective sleep outcomes in the adolescent brain cognitive development study. (Harriman et al., 2024)	Fitbit; Sleep Duration	Sociodemographic disparities.	Health Disparities	N = 11,187; n = 5901 with available Fitbit data.	Fitbit-derived sleep negatively associated with exposure to legal problems, maternal hardship, and housing insecurity, and membership in LCA-derived group experiencing elevated interpersonal stressors.
Social epidemiology of Fitbit daily steps in early adolescence. (Nagata et al., 2023)	Fitbit; Step Count	Sociodemographic differences.	Health Disparities	N = 6038	Greater Fitbit-derived step counts associated with male sex, Black/African American race, and heterosexual sexual orientation. Participants with married parents and from high-income households also recorded higher daily step counts.
Disparities in sleep duration among American children: effects of race and ethnicity, income, age, and sex. (Giddens et al., 2022)	Fitbit; Sleep Duration	Sociodemographic disparities.	Health Disparities	N = 4207	Lower Fitbit-derived sleep duration observed in Black/ African American, male, and older participants. Low household income and higher body mass index also predictive of shorter sleep duration.
Race, Ethnicity, and Sleep in US Children. (Wang et al.; 2024)	Fitbit; Sleep Duration, timing, variability	Racial/Ethnic differences.	Health Disparities	N = 3868	Later bedtimes, shorter and more variable sleep duration observed in non-white race and multiracial participants, all with small effect sizes ($\beta = 0.04–0.11$)
Gender diversity and daily steps: Findings from the Adolescent Brain Cognitive Development Study. (Nagata et al., 2025)	Fitbit; Step Count	Sociodemographic Disparities.	Health Disparities	N = 6460	Transgender identity and reported gender dissatisfaction associated with reduced step counts.
Using explainable machine learning and Fitbit data to investigate predictors of adolescent obesity. (Kiss et al., 2024)	Fitbit; Multiple sleep and physical activity metrics	Obesity	Machine Learning	N = 2971; n = 401 with obesity	Obesity significantly associated with later bedtimes, reduced sleep duration, variable bed/ wake times, low step counts, high sleeping and daytime resting heart rates. In Explainable Boosting Machine (EBM) models, a combination of Fitbit metrics and sociodemographic features yielded acceptable classification accuracy (Mean AUC = 0.726) in predicting obesity status.
Machine Learning-Based Prediction of Attention-Deficit/Hyperactivity Disorder and Sleep Problems With Wearable Data in Children. (Kim et al., 2023)	Fitbit Multiple sleep and physical activity metrics	ADHD, Sleep problems	Machine Learning	N = 1090; n = 79 with ADHD	ADHD (79 cases vs. 1011 controls) associated with shorter sleep, higher heart rate, greater energy expenditure, increased activity level. Sleep problems (68 cases vs. 3346 controls) associated with reduced REM sleep, more frequent nighttime waking, poorer sleep quality, reduced physical activity and step counts. Acceptable classification accuracy achieved for ADHD (AUC=0.798) and sleep problems (0.737) from combination of Fitbit metrics, physical features, and participant sex.
Unlocking the potential of wearable technology: Fitbit-derived measures for predicting ADHD in adolescents. (Rahman, 2025)	Fitbit; multiple physical activity metrics	ADHD	Machine Learning	N = 450; n = 225 with ADHD	In subsample of 450 participants (225 with ADHD and 225 demographically matched controls), ADHD diagnosis associated with higher and more variable resting heart rate, greater daytime and reduced nighttime sedentary behavior,

(*continued on next page*)

Table 1 (*continued*)

Article Title/Citation	Sensor; Measure	Study Outcome	Grouping	Sample Size	Primary Sensor-Based Findings
					and greater variability in sedentary behavior.
Digital phenotyping from wearables using AI characterizes psychiatric disorders and identifies genetic associations. (Liu et al., 2025)	Fitbit; multiple static and time-varying sleep and physical activity metrics	ADHD, Anxiety	Machine Learning	ADHD: N = 1953 (216 cases) Anxiety: N = 2403 (666 cases) GWAS: N = 1191	A combination of static and time-varying Fitbit sleep and physical activity metrics paired with demographic, clinical, and neurocognitive measures yielded highly accurate prediction of ADHD (N = 216 cases, 1737 controls; AUC = 0.89) and acceptable prediction of anxiety (N = 666 cases, 1737 controls; AUC = 0.71) in gradient-boosted machine learning models (XGBoost). Inclusion of Fitbit metrics yielded statistically significant improvement in predictive accuracy for both outcomes. Multivariate digital phenotyping scores for ADHD and anxiety were significantly related to novel genetic loci in relatively small genome-wide association analyses.
Association of Demographic and Socioeconomic Indicators With the Use of Wearable Devices Among Children. (Kim et al., 2023)	Fitbit; device wear time	Study Participation and Attrition	Measurement Validity	N = 10,414	Black ABCD participants and participants from low-SES households were less likely to participate in wearables trial and, amongst trial participants, registered significantly lower wear time and fewer wear days compared to White or high-income participants.
Accuracy of parent-reported sleep duration among adolescents assessed using accelerometry. (Turan et al., 2025)	Fitbit; sleep duration	Concordance between parent-reported and Fitbit-recorded sleep duration.	Measurement Validity	N = 5855	Parent-report significantly overestimated youth sleep compared to Fitbit measures. Few participants (0.8%) recorded 9 or more hours of sleep per night via Fitbit, the recommended total for their age group.
Assessment of Sleep Measures and their agreement: Youth-Reported, Caregiver-Reported, and Fitbit-Derived Data in a Large Early Adolescent Cohort. (Kiss et al., 2025)	Fitbit; sleep duration, bed/wake times, onset latency, social jetlag	Concordance between youth-reported and Fitbit-recorded sleep metrics.	Measurement Validity	N = 3803	Adolescent report most concordant with Fitbit metrics for bed times (ICC =.460), modestly concordant for sleep duration (ICC = 0.186) and no relationship to nightttime waking (ICC=-0.015) or sleep onset latency (ICC=-0.009). Agreement between Fitbit and youth-report weakest for youth reporting less than 7 h of nightly sleep.
Examining the Agreement Between Subjective and Objective Measures of Sleep: A Comparison of Munich Chronotype Questionnaire and Fitbit-Derived Sleep Metrics. (Rohr et al., 2025)	Fitbit; Sleep duration, sleep midpoint, social jetlag	Convergent validity of Fitbit Sleep and Munich Chronotype Questionnaire (MCTQ)	Measurement Validity	N = 5252	Fitbit and MCTQ estimates weakly-to-moderately correlated for sleep duration (*r = .15-.21), social jetlag (0.12–0.16), and sleep midpoint (r = 0.37–0.42)*. Fitbit classified greater proportion of participants (63.2%) as attaining insufficient sleep compared to MCTQ (39.5%).
Recommendations for Identifying Valid Wear for Consumer-Level Wrist-Worn Activity Trackers and Acceptability of Extended Device Deployment in Children. (Wing et al., 2022)	Fitbit; Step count, heart rate, physical activity, wear time	Distinguishing valid and invalid Fitbit wear time, trial acceptability	Measurement Validity	N = 137	Nearly all youth reported comfort with Fitbit device and willingness to participate in future Fitbit trial (98%). After data processing, 77.7% of recorded minutes were retained as valid, with usable data available from 89% participants.
Passive Sensing of Preteens' Smartphone Use: An	EARS; Smartphone application use duration,	Smartphone use descriptive statistics, congruence with	Measurement Validity; Screen Use	N = 67	Mean passively sensed smartphone use duration

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Table 1 (*continued*)

Article Title/Citation	Sensor; Measure	Study Outcome	Grouping	Sample Size	Primary Sensor-Based Findings
Adolescent Brain Cognitive Development (ABCD) Cohort Substudy. (Wade et al., 2021)	time of day, category, specific application use	youth and parent-reported smartphone use.			moderately correlated with youth self-report (*r = .49) and weakly correlated with parent-report (r = .06 −.22).* Passively sensed use averaged 3 h, 45 min per day, significantly greater than youth's pre-trial estimates (2 h, 45 m) though consistent with youth retrospective report of use during the passive sensing study (3 h 28 m).
Passively sensing smartphone use in teens with rates of use by sex and across operating systems. (Alexander et al., 2024)	EARS; Smartphone application usage timing and duration, keystroke counts, application use category, specific applications used	Smartphone Use descriptive statistics, psychometric properties, convergence with youth self-report.	Measurement Validity; Screen Use	N = 1463 (501 Android/ 962 iOS device users)	Smartphone application use and keystroke counts exhibited high internal consistency (Cronbach's α =.96 −.98) and moderate stability (ICCs =.51 −.58). Keystrokes and application use moderately correlated (*r = .33).* Passively measured app use duration 113 min greater on average than youth estimate (298 m vs, 185 m per day) Youth self report weakly correlated with keystroke count (r = .13–21) and moderately correlated with application use duration (*r = .35).*
Examining measurement discrepancies in adolescent screen media activity with insights from the ABCD study. (Zhao et al., 2025).	EARS; Smartphone usage duration, social media, gaming, specific application use	Convergence of youth self-reported and EARS smartphone use measures; associations between EARS and mental health and neurocognitive outcomes.	Measurement Validity; Screen Use; Mental Health	N = 495	Total smartphone use averaged 4.89 h per day and was moderately correlated with youth self-reported total screen time (*r = .45).* Mean youth-reported social media use (1.63 h/day) consistent with passively sensed use (1.64 h/ day) over the same period. Passive smartphone use associated with youth-reported externalizing (β = 0.07) and social media addiction (β = 0.06).
The association of objectively and subjectively measured modifiable lifestyle factors with internalizing problems: the role of genetic confounding and shared method variance bias. (Zhang et al., 2025)	Fitbit; Sleep duration, exercise intensity and duration	Child Behavior Checklist (CBCL) internalizing scale.	Mental Health	N = 2772 (Physical activity) N = 2253 (Sleep) N = 2084 (Both sleep and physical activity)	Greater sleep duration (β = −0.10) and greater moderate to vigorous physical activity (β = −0.05) measured at the two-year follow up visit prospectively predicted reduced internalizing problems one year later. Litte evidence found for genetic confounding of relationships in polygenic scoring analyses, though the effect of moderate to vigorous physical exercise, while numerically similar, was no longer statistically significant after partially controlling for genetic confounding.
Genetic influence and neural pathways underlying the dose-response relationships between wearable-measured physical activity and mental health in adolescence. (Yu et al., 2025)	Fitbit; Daily step counts, physical activity duration, intensity	Child Behavior Checklist (CBCL) scales.	Mental Health	N = 10,414	Greater daily step counts, longer moderate and vigorous physical activity duration associated with lower CBCL internalizing, thought, and total problems scores. Greater light physical activity associated with greater CBCL internalizing and thought scales. Nonlinear models supported U-shaped dose-response relationships between moderate and vigorous physical activity with CBCL behavioral health scales.

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Table 1 (*continued*)

Article Title/Citation	Sensor; Measure	Study Outcome	Grouping	Sample Size	Primary Sensor-Based Findings
Physical and Mental Health in Adolescence: Novel Insights from a transdiagnostic examination of Fitbit data in the ABCD Study. (Damme et al., 2024).	Fitbit; resting heart rate, physical activity intensity/ duration	Child Behavior Checklist (CBCL) Internalizing and Externalizing symptoms. Psychosis-like experiences.	Mental Health	N = 5007	Higher resting heart rate associated with greater internalizing symptoms. Greater time spent sedentary associated with greater endorsement of psychosis-like experiences. Greater duration of moderate-to-vigorous physical exercise associated with lower internalizing and psychosis-like symptoms.
Concurrent and prospective associations between Fitbit wearable-derived RDoC arousal and regulatory constructs and adolescent internalizing symptoms. (Nelson et al., 2022)	Fitbit; resting heart rate, daily step count, sleep duration	Physical Activity, Heart Rate	Child Behavior Checklist Internalizing Symptoms	N = 5686 n = 143 (prospective analyses)	Cross-sectionally, greater internalizing symptoms were associated with higher resting heart rate, lower and less variable daily step counts, and more variable nightly sleep duration. Baseline Fitbit metrics did not prospectively predict internalizing symptoms two years later. Baseline internalizing weakly predicted future step counts and baseline resting heart rate weakly predicted future step counts and step count variability.
Latent Profiles of Sleep Patterns in Early Adolescence: Associations With Behavioral Health Risk. (Zhang et al., 2024)	Fitbit; Latent sleep profiles derived from multiple Fitbit metrics	Latent sleep profile classification, association with behavioral health symptoms	Mental Health	N = 3326	Four latent sleep profiles identified from Fitbit sleep metrics: average sleep; high duration/ high wakefulness; high efficiency/ low duration/ low wakefulness; low duration/ low efficiency. Low duration/ low efficiency membership associated with greater rule breaking behaviors, attention problems, and social problems compared to average sleepers. High duration/high wakefulness associated with lower rule breaking and attention problems compared to low duration/low efficiency group.
The relationship between sleep and problem behaviors in children with attention-deficit/hyperactivity disorder. (Ghanim et al., 2025)	Fitbit; Sleep duration, onset latency, nighttime awakenings, sleep stage duration	Sleep	Child Behavior Checklist (CBCL) Subscales, ADHD case/control status.	N = 424 (212 ADHD cases and sex-matched controls)	No significant differences between ADHD cases and controls for Fitbit sleep measures after Bonferroni correction. Greater sleep onset latency, contrary to expectations, modestly predicted fewer attention and aggression problems.
Associations Between Adverse Childhood Experiences and Early Adolescent Physical Activity in the United States. (Al-shoaibi et al., 2024)	Fitbit; Average daily step count		Mental Health; Health Disparities	N = 7046	Daily step counts modestly lower in participants reporting adverse childhood experiences, emotional abuse, physical neglect, household mental illness, or parental divorce.
Adverse Childhood Experiences and Accelerometer-Measured Physical Activity and Sleep in Preadolescents (Lewis-de los Angeles, 2024)	Fitbit; Step count, physical activity duration/ intensity, sleep duration, REM sleep duration	Adverse childhood experiences	Mental Health; Health Disparities	N = 6227 (Physical activity) N = 4151 (Sleep)	Presence and quantity of adverse childhood experiences both modestly associated with fewer daily steps, reduced moderate-to-vigorous physical activity duration, and shorter and shorter REM sleep duration. Presence of two or more adverse childhood experiences also modestly associated with reduced total sleep duration. The paper asserts support for a dose-response relationship, though this is not explicitly tested.

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Table 1 (*continued*)

Article Title/Citation	Sensor; Measure	Study Outcome	Grouping	Sample Size	Primary Sensor-Based Findings
Physical Activity as a Buffer in the Association Between Perceived Ethnic-Racial Discrimination and Latinx Adolescent Mental Health (Zhang & Halgunseth, 2024)	Fitbit; Physical activity intensity/ duration	Child Behavior Checklist Internalizing Problems	Mental Health; Health Disparities	N = 1392	In subsample of Hispanic/Latino adolescents, no significant associations observed between duration of moderate-to-vigorous physical activity and internalizing symptoms, though the association between internalizing symptoms and perceived discrimination was modestly weaker in adolescents with longer duration moderate-to-vigorous physical activity.
Sleep as a Protective Factor: Multiple Forms of Discrimination and Substance Use Intention Among Racially and Ethnically Minoritized United States Youth. (Wang et al., 2024).	Fitbit; Sleep duration and variability	Substance use, intentions discrimination experiences	Mental Health; Health Disparities	N = 3495	The association between experiencing discrimination and future substance use intentions attenuated in adolescents with greater weekday sleep duration ($\beta = -0.05$). Weekday sleep duration was interactive with sleep variability, such that longer duration predicted reduced substance use intentions only in youth with more consistent weekday sleep ($\beta = -0.06$).
Fitbit-Measured Sleep Duration in Young Adolescents is Associated with Functional Connectivity in Attentional, Executive Control, Memory, and Sensory Networks. (Turan et al., 2025)	Fitbit; Sleep Duration	Resting state functional connectivity.	Neuroimaging	N = 3799	Fitbit sleep duration significantly associated with 36/78 pairwise functional network connections after FDR-correction, with stronger associations observed in girls than boys (38 vs. 11 pairwise significant pairs). Longer sleep was modestly associated with stronger within-network connectivity in the cingulo-opercular, dorsal-attention, default-mode, retrosplenial-temporal, and visual networks and weaker within-sensorimotor network connectivity.
Functional brain connectivity predicts sleep duration in youth and adults. (Mummaneni et al., 2023)	Fitbit; Sleep duration	Whole-brain functional connectivity (resting state and task-based)	Neuroimaging	n = 1274 (fMRI resting state) n = 786 (fMRI N-back task)	Whole-brain functional connectivity significantly predictive of sleep duration in both N-back task and resting state data (*r = .143 −.238)*, with resting state prediction significantly stronger than task-based prediction.
Getting a Good Night's Sleep: Associations Between Sleep Duration and Parent-Reported Sleep Quality on Default Mode Network Connectivity in Youth. (Hehr et al., 2023)	Fitbit; Sleep duration, bed/wake times, wake after sleep onset (WASO)	Resting state functional connectivity between default mode (DMN), dorsal attention, frontoparietal, and salience networks.	Neuroimaging	N = 3798	Shorter sleep duration was weakly predictive of within-network DMN connectivity, greater between-network connectivity between the DMN and frontoparietal and dorsal attention networks. Wake after sleep onset was interactive with sleep duration, such that WASO predicted greater DMN dorsal attention connectivity only in youth sleeping less than 7.5 h each night.
Effects of sleep-corrected social jetlag on measures of mental health, cognitive ability, and brain functional connectivity in early adolescence. (Yang et al., 2023)	Fitbit, Sleep-corrected social jetlag	Child Behavior Checklist (CBCL) scales, Psychosis experiences, NIH Toolbox Neurocognitive measures, School performance, Resting State Functional Connectivity, Gray matter volume	Neuroimaging; Cognition	N = 6335	Fitbit-derived sleep-corrected social jetlag weakly correlated with questionnaire-derived social jetlag ($r = .23$). Small-moderate associations observed with lower crystallized intelligence and poorer grades (Cohen's d =.21 −.30) and weak associations observed with 33/306 functional connectivity measures.

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Article Title/Citation	Sensor; Measure	Study Outcome	Grouping	Sample Size	Primary Sensor-Based Findings
Neural correlates of device-based sleep characteristics in adolescents. (Ma et al., 2025)	Fitbit; 18 metrics related to sleep duration, fragmentation, heart rate, timing, and sleep onset latency	Brain region structure and volume measures, resting state functional connectivity. Neurocognitive outcomes	Neuroimaging; Cognition	N = 3222	PCA-derived later-onset/shorter sleep dimension associated with reduced cortical-subcortical connectivity. A higher heart rate/shorter light sleep factor was associated with lower regional brain volumes and reduced connectivity. "Sleep-informed biotypes" derived from latent class analysis of sleep-brain dimensions exhibited differences in neurocognitive performance, with an earlier, longer sleep group exhibiting significantly stronger crystallized intelligence and reading achievement scores.
Somatomotor Disconnection Links Sleep Duration With Socioeconomic Context, Screen Time, Cognition, and Psychopathology. (Cleanthis et al., 2025)	Fitbit; Sleep duration, latent multimethod sleep duration factor variable	fMRI resting state functional network connectivity.	Neuroimaging; Cognition; Mental Health	N = 3037	A multimodal sleep duration composite measure, which included Fitbit sensor metrics, was modestly associated with reduced somatomotor network connectivity and integration.
Latent Default Mode Network Connectivity Patterns: Associations With Sleep Health and Adolescent Psychopathology. (Zhang et al., 2025)	Fitbit; Sleep duration, efficiency	Latent default mode network functional connectivity, Child Behavior Checklist internalizing and externalizing symptoms	Neuroimaging; Mental Health	N = 2811	Fitbit sleep duration and sleep efficiency metrics were modestly predictive of membership in one of four latent class analysis-derived profiles characterizing within-network and between-network resting-state functional connectivity for the default mode network.
Changes in Rest-Activity Rhythms in Adolescents as They Age: Associations With Brain and Behavioral Changes in the ABCD Study. (Zhang et al., 2025)	Fitbit; Sleep duration and timing; Daily step count	Developmental changes in sleep and physical activity, associations with demographic, neuroimaging, and behavioral measures	Neuroimaging; Mental Health	N = 963	Average sleep duration 23 min lower, sleep onset 57 min later, and average step count reduced by 1379 steps/day from two-year to four-year follow up. Reduced sleep duration predicted altered resting-state functional connectivity between subcortical, sensorimotor, cingulo-opercular, visual, and auditory networks. Delayed sleep onset associated with changes in subcortical gray matter. Behaviorally, delayed sleep modestly associated with poorer grades and increased parent-reported impulsiveness and behavior problems.
Sports participation & childhood neurocognitive development. (Tan et al., 2025)	Fitbit; Metabolic equivalents, Step count	Sports Participation	Neuroimaging; Physical Health	N = 3481	Fitbit METs and weekly step counts did not significantly differ by participation in youth sports.
Lower daily steps among U.S. adolescents during the COVID-19 pandemic: Objective findings from the Adolescent Brain Cognitive Development Study. (Nagata et al., 2023)	Fitbit; Daily Step Count	Changes in physical activity during COVID-19 pandemic	Physical Health	N = 4955	Participants assessed during the COVID-19 pandemic recorded 20.8% fewer steps per day compared to participants assessed before the pandemic's onset.
Physical Activity and Sedentary Time Among U.S. Adolescents Before and During COVID-19: Findings From a Large Cohort Study. (Hunt et al., 2024)	Fitbit; Daily step count, physical activity intensity/ duration	Changes in physical activity during COVID-19 pandemic	Physical Health	N = 7179	Participants assessed during the school year and after the COVID-19 pandemic recorded significantly fewer daily steps. Fewer minutes of moderate-to-vigorous physical activity, and more minutes spent sedentary than participants assessed during the school year and prior to the COVID-19 pandemic. Pandemic-related declines were also observed for daily steps and

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Article Title/Citation	Sensor; Measure	Study Outcome	Grouping	Sample Size	Primary Sensor-Based Findings
					moderate-to-vigorous physical activity for youth assessed during the Summer months, where declines were also significantly stronger for boys.
Impact of Environmental Noise and Sleep Health on Pediatric Hypertension Incidence: ABCD Study. (De Moraes et al., 2025)	Fitbit; Sleep duration, REM-sleep duration, Step Count, Physical activity duration	Sleep	Physical Health	N = 4721 (Baseline) n = 3320 (Fitbit data, outcomes, and covariates available)	Healthy sleep duration at two-year follow up was associated with lower hypertension risk two years later (RR = 0.63). No significant interaction between sleep duration and environmental noise on hypertension risk.
The influence of pubertal development on early adolescent sleep and changes in family functioning. (Peltz et al., 2023)	Fitbit; Sleep duration	Adolescent-reported family conflict. Interaction with pubertal timing.	Physical Health	N = 4682	Associations between earlier pubertal timing and increased family conflict mediated by changes in sleep duration and variability. More advanced pubertal stage was weakly associated with shorter and more variable sleep duration, which in turn weakly predicted increases in family conflict.
Exploring the associations between the presence, characteristics, and biopsychosocial covariates of pain and lifetime depression in adolescents: A cross-sectional ABCD study analysis. (Tagliaferri, 2025)	Fitbit; Daily step counts, sedentary behavior, time physically active; duration of light, deep, REM, and total sleep.	Lifetime depressive disorder diagnosis, depression/pain status.	Physical Health; Mental Health	N = 5211	No significant associations between Fitbit sleep or physical activity metrics, depression, or pain status (i.e., all $p > .05$ after FDR correction)
Association of Physical Activity and Screen Time With Body Mass Index Among US Adolescents. (Nagata et al., 2023)	Fitbit; Daily step count	Body mass index, overweight/ obesity status	Physical Health; Screen Use	N = 5797	Low daily steps (less than 6000 steps/day) and medium steps (between 6000 and 12,000 steps/day) associated with significantly greater overweight or obesity risk than high daily step group (greater than 12,000 steps/day). The effect of youth-reported screen use on BMI was moderated by daily step count, such that high screen use was associated with a greater increase in BMI in the low step count group compared to the medium and high step count groups.
Association of physical activity and screen time with cardiovascular disease risk in the Adolescent Brain Cognitive Development Study. (Nagata et al., 2024)	Fitbit; Daily step count	Blood pressure percentile, Cholesterol, A1c.	Physical Health; Screen Use	N = 4718	Greater daily steps associated with lower diastolic blood pressure and lower total and HDL cholesterol.
Prospective associations between sleep, sensation-seeking, and mature screen usage in early adolescents: findings from the Adolescent Brain Cognitive Development study. (Zhang et al., 2025)	Fitbit; Sleep duration, bed time, social jetlag.	Sleep	Mature videogame playing, R-rated movie watching.	N = 3687 n = 992 (bidirectional analyses)	Shorter sleep duration, later and more variable bedtimes, and social jetlag were all weakly prospectively associated with greater mature video gaming and R-rated movie watching one year later. Bidirectional analyses further found reciprocal associations between mature screen use, bedtime, bedtime variability, and social jetlag. Sleep duration moderated association between sensation seeking and R-rated movie watching.
Adolescent smartphone use, sleep, and physical activity: daily associations between sensor-based measures in the adolescent brain cognitive development (ABCD) study. (Alexander et al., 2025)	EARS, Fitbit; Smartphone application and keyboard use duration and timing, Sleep duration, Bed/wake times, Duration/ intensity of physical activity	Within/between person associations between smartphone use, sleep, and physical activity.	Screen Use	*N = 791*	Smartphone use during the day time and evening associated with fewer daily steps, less time spent exercising, and more time sedentary (β = −.21 –.07). Smartphone use late at night associated with significantly

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Article Title/Citation	Sensor; Measure	Study Outcome	Grouping	Sample Size	Primary Sensor-Based Findings
					shorter sleep duration, delayed sleep onset and waking, reduced sleeps, and increased sedentary time (β = −.26 –.27). Relationships generally persisted within person, indicating that adolescents slept less or later and were less active on days when they used their phones more than usual.

finding Fitbit data collection comfortable (98%) and enjoyable (87%) and expressed interest in future data collection efforts (98%). Pre-processing steps included removal of minute-level observations with invalid heart rate reads and establishing minimum wear times for computation of daily and weekly-level data.

Other papers used the complete Fitbit dataset available from ABCD participants to consider the broader generalizability of Fitbit and correspondence to self-report within ABCD. For instance, Kim et al. (2023) identified sociodemographic factors associated with willingness to participate and protocol adherence, comparing the 7424 participants who shared any wearable data with the 2900 who did not. Findings indicated participation rates, protocol adherence, and data quality were significantly lower for Black participants and participants from lower-income households, highlighting considerations for equity, generalizability, and statistical biases due to selection effects. Accuracy of self-reported sleep metrics relative to Fitbit has been considered from several angles. Two studies comparing Fitbit and parent-reported measures of sleep duration, obtained via the Sleep Disturbance Scale for Children (SDSC), each identified substantial parent overreporting of adolescent sleep duration by over one hour (Kiss et al., 2025; Turan et al., 2025). Comparisons between Fitbit and youth self-reported sleep (via the Munich Chronotype Questionnaire) also found youth reported significantly longer sleep duration and earlier sleep onset compared to Fitbit measures (Kiss et al., 2025; Rohr et al., 2025). Correlations between youth-reported and Fitbit-derived sleep duration, sleep midpoint, and social jetlag were modest (*rs*=0.152–0.364) though, notably, these were not collected concurrently; self-reported sleep measures were completed prior to Fitbit data collection.

6.2. Smartphone sensor data

Initial efforts to validate ABCD's smartphone sensor data, which reflect device-based estimates of participant application and keyboard usage (for Android devices) or keyboard usage collected via an application-based keyboard overlay (for Apple iOS devices) included the computation of psychometric statistics, comparisons to self-report-based measures of smartphone use, and comparisons to expected day/night and weekday/weekend usage patterns.

Wade et al. (2021) investigated validity and acceptability of sensor-based data collection in a small ABCD subsample (n = 67) of early adolescents (ages 11–12) with Android devices. This study found that a majority of adolescents (69%) and their parents (85%) reported interest in ongoing participation in sensor-based smartphone data collection. Youth pre-trial estimates of daily smartphone use underestimated sensor-based estimates by an average of one hour while youth post-study estimates of smartphone use during the trial were moderately-highly correlated with sensor-based estimates (r = .49). Parent reported estimates of smartphone use were largely inaccurate, with weak and non-significant associations with device-measured use (r = .10).

Further validation efforts of passive smartphone data collected at the 4-year follow-up visit (n = 1463; ages 13–14) were conducted in Alexander et al. (2024). This study found moderate inter-day variability (ICCs=.51 −.58) and high internal consistency (α=.96 −.98) of application and keystroke measures. Total recorded use on weekends and weekdays did not differ significantly while intraday usage patterns indicated greatest use in the afternoons and evenings and lowest use late at night and early in the morning. Youth underreported smartphone use by an average of 1.9 h per day, though a later analysis by Zhao et al. (2025) found that sensor-based measures of smartphone use were similar to youth's self-reported total screen use during the three-week sensing trial (n = 495). Self-reported and passively sensed smartphone use, which were not measured concurrently, were moderately correlated with one another (r = .35), though correlations with keyboard usage were modest (*rs*=.13 −.21). Further, Zhao et al. (2025) assessed associations between self-reported and passively sensed smartphone use further stratified by application category, finding that correlations were especially strong for social media use (r = .64) and weak and non-significant for video games (r = .08*)*, likely reflecting gaming using other devices.

7. Machine learning prediction applications

Recent studies using wearable data from the ABCD cohort demonstrate that machine learning (ML) can extract clinically meaningful digital phenotypes predictive of psychiatric and behavioral outcomes. These investigations reflect a shift from single-modality feature extraction toward interpretable, multimodal frameworks that integrate physiology, behavior, and environment to model health risk at scale. One such study, Liu et al. (2025), applied an interpretable deep learning model (Xception) to over 250 Fitbit-derived features for classifying psychiatric disorders through 4-year follow-up (ages 9–15). Their approach achieved high predictive performance for externalizing (ADHD; AUROC = 0.89) and internalizing (anxiety) disorders (AUROC = 0.83), surpassing covariate-based and neuroimaging-derived models. Interpretability analyses localized predictive temporal windows, including daytime heart rate for ADHD and nocturnal sleep for anxiety, and integration with genome-wide association studies (GWAS) identified 16 novel loci and 37 psychiatric-associated genes, linking wearable-derived physiology to genetic mechanisms. Rahman et al. (2025) similarly demonstrated that Fitbit-based measures of sedentary time, resting heart rate, and energy expenditure accurately predicted ADHD diagnoses to matched controls at ages 13–14 (AUC = 0.95) using Random Forest classifiers, outperforming neuroimaging benchmarks and highlighting the scalability of low-cost, passive data for behavioral health prediction. Kim et al. (2023) extended this approach using 21 days of Fitbit data and 64 circadian rhythm features in a Light Gradient Boosting Machine (LightGBM) model (AUC = 0.798) when ABCD participants were 11–12. Heart rate metrics (MESOR and maximum HR) were the strongest predictors of ADHD diagnosis and sleep problems, and high negative predictive values (NPV > 0.97) suggest promise for early detection and screening. Beyond psychiatric traits, Kiss et al. (2024) applied Explainable Boosting Machines (EBMs) to predict adolescent obesity at ages 11–12 (AUC = 0.726) and uncovered

non-linear risk thresholds and interactions, particularly between sleeping heart rate and socioeconomic adversity, highlighting the role of context-sensitive, interpretable modeling. Collectively, these studies underscore heart rate and sleep features as consistent physiological markers across domains and illustrate how explainable, equity-aware ML can advance scalable digital phenotyping.

8. Mental health

Fifteen articles were identified which examined associations between ABCD passive sensing measures and adolescent mental health outcomes. A single article examined associations between passively sensed smartphone use and mental health while the remaining fourteen examined associations with Fitbit-based measures of sleep and physical activity. All identified articles reported significant mental health associations, though effect sizes were generally small in magnitude.

Three papers identified modest associations between Fitbit-derived measures of physical activity, including resting heart rate, step count, and activity intensity, and symptoms of internalizing psychopathology at ages 11–12 (Damme et al., 2024; Nelson et al., 2022; Yu et al., 2025). One such paper included longitudinal data from a subsample of participants (n = 143) and additionally reported prospective associations between internalizing symptoms and slightly reduced step counts one year later (Nelson et al., 2022). Another of these studies further reported modest associations between psychosis-like experiences and greater sedentary time and reduced time spent exercising (Damme et al., 2024). The third study included both linear and non-linear modeling and polygenic risk score analyses (Yu et al., 2025). In addition to linear internalizing relationships, this study reported linear relationships between step count and physical activity with thought disorder symptoms and total psychopathology symptoms, nonlinear relationships between total mental health symptoms and physical activity (with daily exercise between 90 and 120 min per day predicting the lowest symptom counts), and associations between polygenic risk indices for psychiatric traits (specifically anxiety, autism spectrum disorder, and schizophrenia) with Fitbit-based step count and physical activity measures.

Five studies examined associations between Fitbit-derived sleep measures and mental health outcomes. In a study of youth ages 10–13 with ADHD (n = 212) compared to matched controls (Ghanim et al., 2025), findings indicated no difference between those with ADHD and controls in Fitbit sleep metrics. Further, Fitbit sleep metrics did not relate to mental health scores on the CBCL, though parent-reported sleep problems were associated with parent-reported mental health scores on the Childhood Behavior Checklist (CBCL; Achenbach, 2015). In another study of Fitbit sleep metrics at ages 10–13, latent profile analyses (LPA) were conducted to categorize sleep characteristics. Participants whose metrics were classified as low sleep duration and low sleep efficiency demonstrated the highest parent-reported CBCL scores on subscales for attention, social problems, and rule-breaking (Zhang et al., 2024). A separate analysis with the same LPA structure (n = 2811) investigated resting state fMRI in the default mode network, functional connectivity, and Fitbit-derived sleep duration and efficiency (Zhang et al., 2025). Youth who had lower sleep efficiency were more likely to be classified as having high connectivity within the DMN but low connectivity between DMN and other networks. Youth with this pattern also had higher externalizing symptoms reported one year later (Zhang et al., 2025). Finally, Fitbit data collected between ABCD Study 2-year (ages 11–12) and 4-year (ages 13–14; n = 963) follow-up found that later sleep onset was modestly associated with increased impulsivity and mood problems (Zhang et al., 2025).

Fitbit data within ABCD are also being used for other cross-cutting analyses. Consistent with prior studies, Zhang et al., (2025) used 2-year follow-up data to assess sleep and mental health (n = 2772). Findings indicated that self-reported and Fitbit device-measured increase in sleep duration and physical activity both relate to decreased internalizing symptoms. Further, genetic modeling showed self-report effects were largely explained (81% variance) by shared genetic variance and method bias, while wearable effects did not demonstrate genetic confounding of relationships. This indicates the potential of passive sensing to sensitively detect small effects with reduced concerns for some biases. On the other hand, sensors may not always identify relationships. For instance, in a study of ABCD participants at 2-year follow up (ages 11–12), while reporting any pain experience was associated with lifetime depression (Tagliaferri et al., 2025), pain did not predict physical activity as measured by Fitbit. As many studies have been done at ages ~11–12 in ABCD, a broader age range needs to be considered to assess sensitivity and specificity for these clinical relationships over time.

For passively sensed smartphone use and mental health, a study using 4-year follow-up data (ages 13–14, n = 495) found that when assessing self-report and objective passive sensing metrics and clinical outcomes, both predicted increased externalizing symptoms, internalizing symptoms, and social media addiction, all with small effects (Zhao et al., 2025). Taken together with the Fitbit literature, sensor-based cardiovascular, movement, and smartphone data show clear, though often modest in effect size, concurrent associations with mental health functioning. More longitudinal analyses with larger segments of the cohort are needed to assess long-term outcomes.

Across studies, then, it is clear that sensor-based measures are feasible, sensitive tools for detecting physiological and psychiatric relationships. This is particularly true for Fitbit-based measures, but there is also initial evidence for smartphone sensors. At the same time, the clinical utility of these relationships is not currently clear, with modest effect sizes, and findings are largely restricted to ages 11–12 due to the data available within the current data releases. Benefits of sensor data in identifying mental health relationships include potential to reduce bias, including from self-report and other confounds, including potentially genetics. As more longitudinal and comprehensive analyses are conducted, optimal use of sensors relative to self-report will likely become clearer and help accomplish ABCD's aims regarding assessing personal, modifiable factors which influence adolescent mental and physical health development.

9. Physical health

Eight studies were identified that utilized Fitbit sensor measures to investigate physical health outcomes in ABCD participants. Two studies examined changes in physical activity during the COVID-19 pandemic. One study found that participants assessed in the first year of the pandemic logged, on average, 20.8% fewer daily steps compared with participants assessed before the pandemic (Nagata, Yu, et al., 2023). Another found that, compared to pre-pandemic levels, adolescents took an average of 1782 fewer steps per day, engaged in 6 fewer minutes of moderate-to-vigorous exercise, and were sedentary 30 min longer each day (Hunt et al., 2024).

Several studies have investigated associations between Fitbit derived measures of both physical activity and sleep with cardiovascular risk outcomes, including blood pressure and indicators of obesity. One study found significant associations between lower daily step counts and higher diastolic blood pressure (Nagata et al., 2024) and another reported associations between lower step counts, higher body mass index, and increased risk for obesity (Nagata, Smith, et al., 2023). Another study, Kiss et al. (2024), used a machine learning model to investigate associations between multivariate Fitbit-based measures of sleep, physical activity and heart rate to predict adolescent obesity. Key predictors of obesity risk included higher resting and sleeping heart rate, shorter and irregular sleep, and lower step count, with heart rate and sleep measures exhibiting nonlinear associations with obesity risk which further interacted with sociodemographic variables, including race and lower socioeconomic status (Kiss et al., 2024). De Moraes et al. (2024) examined associations between hypertension risk, Fitbit-derived sleep duration, and geocoded environmental noise measures using

longitudinal data from the ABCD study, finding that adolescents who obtained 9 or more hours of sleep per night were protected against increased risk for hypertension regardless of neighborhood noise levels. One article described a study of associations between pubertal development and Fitbit-derived sleep duration and their effects on measures of family functioning (Peltz et al., 2024). Further pubertal development and faster tempo of pubertal development from ABCD baseline (ages 9–11) to two-year follow-up (ages 11–13) were associated with reduced sleep duration and increased nightly sleep variability at two-year follow up, with sleep changes partially mediating increases in conflict with parents and siblings, suggesting sleep as a modifiable risk factor for increased family conflict during pubertal development (Peltz et al., 2024).

Finally, two studies reported null findings between Fitbit's sensor-derived physical activity and sleep measures with the study outcomes. One study found no differences in Fitbit-derived measures of physical activity by adolescent participation in group or individual organized sports activities (Tan et al., 2025). Another, an examination of psychosocial predictors of adolescent pain and depression diagnosis found no differences in Fitbit measures of sleep or physical activity for adolescents reporting past-month pain or comorbid pain and depression compared to adolescents not reporting mental health concerns (Tagliaferri et al., 2025).

Taken together, these studies find significant associations between greater physical activity and sleep health with adolescent physical health, especially cardiovascular health and obesity risk, though associations for other health conditions and behaviors (physical pain and sports participation) in adolescence were not supported.

10. Screen media activity patterns and prediction

Seven articles were identified investigating associations between ABCD sensor measures and digital technology-related health behaviors. Three utilized sensor-based smartphone usage measures, one utilized both Fitbit and smartphone sensor data, and three examined associations between self-reported SMA and Fitbit-based measures of sleep and physical activity.

Beyond validity and acceptability as described above in *Measure Validation*, the three studies described more nuanced patterns of SMA. Wade et al. (2021), conducted at ages 11–12, found participants spent the most time using video streaming apps (e.g., Youtube or Netflix) while Alexander et al. (2024) and Zhao et al. (2025), both conducted at ages 13–15, found that social media and texting applications were most prevalent, with Alexander et al. (2024) reporting that such activities represented a majority of total smartphone use. Alexander et al. (2024) further investigated intraday usage patterns, finding highest rates of use between 4:00 PM and 9:00 PM and lowest between midnight and 5:00 AM, though smartphone use was frequently observed at all hours of the day.

One article examined relationships between sensor-based smartphone use measures and Fitbit-derived measures of sleep and physical activity in ABCD (Alexander et al., 2025). This study found between-person associations between greater smartphone use and less daily step counts and physical activity, and shorter sleep duration. Smartphone use during the daytime and evening exhibited stronger relationships with physical activity while use late at night exhibited stronger relationships with sleep measures. Similar relationships were observed at the within-subject level, such that smartphone use above one's average use predicted same-day reductions in physical activity and sleep duration. Lastly, models of granular within-person hour-by-hour smartphone use effects found that smartphone use prior to sleep onset exhibited little or no impact on sleep-duration, with sleep effects largely driven by late-night smartphone use recorded subsequent to initial sleep onset (Alexander et al., 2025).

The remaining three articles all investigated relationships between Fitbit-based physical activity and sleep measures and self-report measures of SMA. Two articles (Nagata et al., 2023, 2024) examined associations between physical activity, self-reported SMA, and measures of cardiovascular disease risk. Findings indicated that greater physical activity and lower self-reported smartphone use were each independently protective against obesity risk, body mass index, and higher diastolic blood pressure in early adolescents. Nagata et al., (2023) additionally found that, for adolescents with high self-reported SMA, there was no effect of increased step count on body mass index, suggesting protective effects of physical activity on obesity risk may be reduced in adolescents reporting excessive use of SMA. Finally, a longitudinal investigation of relationships between Fitbit sleep measures with engagement with mature screen-based content in ABCD participants identified prospective, bidirectional associations between greater bedtime variability, later bedtimes, shorter sleep duration, and greater social-jetlag with mature-rated gaming and R-rated movie watching (Zhang et al., 2025).

Taken together, these articles demonstrate the utility of passive smartphone sensor data in capturing fine-grained information about adolescent SMA, such as their use of specific smartphone applications and intraday usage patterns, identifying preliminary associations between physical activity, sleep, and SMA patterns, and describing potential interactive effects between physical activity and screen usage impacting adolescent cardiovascular health.

11. Neuroimaging

Converging results across 10 studies linking Fitbit and neuroimaging indicate that irregular timing, curtailed or fragmented sleep, and specific activity exposures are associated with reproducible signatures in white matter and large-scale functional networks. These patterns track with cognitive performance, grades, and mental health, suggesting sleep and circadian health as actionable levers with measurable neural correlates.

A longitudinal analysis of rest–activity rhythms, derived from repeated Fitbit sampling at the 2-year and 4-year follow-ups (i.e., ages 10–13–13–16), found that worsening daily organization with age was associated with alterations in connectivity (linking subcortical regions to sensorimotor and cingulo-opercular systems) and behavioral difficulties, including impulsivity and poorer academic functioning (Zhang et al., 2025). Delays in sleep timing showed stronger links to behavioral problems, whereas shorter sleep duration related to decreased subcortico-cortical (thalamus–cingulo-opercular and amygdala–sensorimotor) connectivity.

Sleep-corrected social jetlag (SJLsc) isolates weekday-weekend timing mismatch after adjusting for sleep duration. In early adolescence (ages 11–12), higher SJLsc was linked to weaker functional coupling between the cingulo-opercular network and the right hippocampus, a memory-relevant circuit (Yang et al., 2023).

Several studies used Fitbit-measured nightly sleep minutes and continuity metrics. Shorter Fitbit-measured sleep at age 11–12 ($N = 3799$) was associated with a multi-network resting state functional magnetic resonance imaging (rs-fMRI) pattern spanning attentional, executive control, memory, and sensory systems, rather than any single-network deficit (Turan et al., 2025). Another analysis using the same wave of data (Hehr et al., 2023) emphasized a default mode network (DMN) signature: shorter sleep, higher wake after sleep onset, and poorer self-reported sleep quality related to weaker within-DMN coupling, blunted DMN–dorsal attention, and DMN–frontoparietal anticorrelations, patterns linked to attention inefficiency in prior work.

Ma et al. (2025) used multivariate dimensions of brain function to define sleep-brain biotypes based on Fitbit data. These two orthogonal sleep-brain dimensions stratified adolescents into three biotypes: Biotype 1, later and shorter sleep with higher heart rate; Biotype 2, intermediate on these axes; Biotype 3, earlier and longer sleep with lower heart rate and superior cognitive performance, from ages 9–14 (Ma et al., 2025). Complementing this, connectome-based models

predicted Fitbit-measured sleep duration in held-out data from both ABCD (ages 11–12) and the Human Connectome Project (HCP; ages 22–35). Critically, the same sleep-predictive connectivity features tracked cognitive task performance, including n-back accuracy and response times, reinforcing that the connectome-sleep signal carries behavioral relevance beyond noise or site effects (Mummaneni et al., 2023).

Sleep effects are not confined to the association cortex. A whole-brain graph-theoretic analysis in > 3000 ABCD youth showed that shorter Fitbit-measured sleep linked to weaker segregation between somatomotor and association networks, a "somatomotor disconnection" pattern that also related to socioenvironmental context. Longer recreational screen time, lower school grades, and lower general cognitive ability co-occurred with greater disconnection, while internalizing symptoms were not elevated; externalizing symptoms increased with greater disconnection (Michael et al., 2025).

Person-centered latent profile modeling of DMN connectivity showed that adolescents ages 11–12 with shorter and less efficient Fitbit device-measured sleep were more likely to exhibit dysregulated DMN profiles. These profiles differed on externalizing symptoms one year later, linking wearable sleep metrics, DMN organization, and behavior (Zhang, Geier, et al., 2025).

Fitbit's sensor-based physical activity measures in ABCD can isolate elements of youth sport participation that relate to brain outcomes. In an analysis of ~11,000 ABCD participants (ages 9–11), youth engaged in organized sports showed superior performance in multiple cognitive domains and greater subcortical gray matter volume and whole-brain white matter integrity (Tan et al., 2025). The analysis incorporated Fitbit-derived metabolic equivalent and step counts, separating structured, social components of sports from raw activity volume.

Despite methodological diversity in these studies, three themes recur. First, device-measured sleep duration and continuity map onto canonical intrinsic networks, especially the DMN and its relations with attention and control systems, with shorter or more fragmented sleep linked to weaker within-DMN coupling and reduced DMN–control anticorrelation (Hehr et al., 2023; Turan et al., 2025). Second, timing regularity matters: SJLsc, a purely temporal construct, shows hippocampal and control-system connectivity differences and indirect links to crystallized cognition and grades, underscoring that stabilizing sleep timing may protect memory-relevant circuits independent of sleep quantity (Yang et al., 2023). Third, effects extend beyond association cortex, encompassing thalamocortical, sensorimotor, and white matter systems, which may mechanistically connect sleep health to attention and mood (Michael et al., 2025; Zhang et al., 2025).

12. Cognition

The use of Fitbit data across ABCD creates a novel opportunity to apply objective metrics to the well-documented relationship between sleep and cognitive task performance. Consistent results across three studies demonstrate that variation in sleep characteristics is associated with reduced performance on cognitive tasks. Fitbit-derived sleep metrics have proven to be insightful and reliable indicators of sleep quality, whether used in conjunction with self-report measures or as independent indices. These data, all collected at 2-year follow-up using Fitbit metrics, allow sleep to be characterized dynamically, incorporating onset (falling asleep), restlessness, waking episodes, duration, and pattern regularity—beyond broad self-reports of falling asleep and wake times.

Michael et al. (2025) demonstrated that shorter sleep duration was associated with reduced connectivity between somatomotor networks and other systems (n = 3037). Follow-up analyses indicated cognitive performance, defined by bifactor modeling of the ABCD cognitive battery, was positively related to somatomotor network connectivity, such that increased connectivity was associated with higher cognitive scores (Michael et al., 2025).

Similarly, inconsistent patterns of sleep onset, waking, and duration between weekdays and weekends, termed 'Social Jetlag', were assessed first by self-report then confirmed with Fitbit-derived metrics (Yang et al., 2023). Social Jetlag was then associated with reduced crystallized intelligence (NIH Toolbox, uncorrected scores).

Finally, Ma et al., (2025) applied sparse canonical correlation analysis (sCCA) to cluster biotypes of sleep quality across onset, restlessness, waking, duration, and heart rate (n = 3222). Biotype 1, characterized by shorter duration, later onset, early waking, and elevated heart rate during sleep, demonstrated lower receptive language, oral reading, and crystallized intelligence compared to the other two biotypes. Biotypes with longer sleep and lower heart rate exhibited better performance. These findings were replicated in the 4-year follow-up sample (n = 1190; Ma et al., 2025).

13. Health disparities

Sensor-data may offer an improved lens to observe healthy equity biases and identify modifiable factors to address health disparities and improve health outcomes. All sensor-based analyses within ABCD which assess the domain of health equity have thus far leveraged the 2-year follow-up data with Fitbit, when youth were approximately 11–12 years-old.

Studies demonstrate clear differences in sleep quality and physical identity based on sociodemographic and identity characteristics. In Giddens et al. (2022), over 7 nights of Fitbit-measured sleep data showed mean total sleep of 7.45 h/night across U.S. children (N = 4207; 7 nights). However, Black children slept 34 min less per night compared with White peers, while children from families earning < $25k annually slept 16 min less than those from higher-income households. Boys and older children also had reduced nightly sleep, largely explained by later bedtimes. Body mass index was an additional predictor of shorter sleep duration. Importantly, neighborhood deprivation and discrimination measures were not significant once adjusted. Similarly, Wang et al. (2024) compared sleep duration and variability across racial and ethnic groups (N = 3868). As in Giddens et al., mean nightly sleep was ~7.5 h, but Black, Latinx, Asian, and multiracial children slept less and had later bedtimes than White peers. Variability in bedtime and risetime was also greater among these groups, with Black youth showing the largest disparities across both duration and variability.

Similar sociodemographic differences in physical activity are noted. Average steps/day were 9096 across the sample. Male adolescents walked 1544 more steps/day than females, while youth from households earning > $200k walked 1003 more steps/day compared with peers from < $25k households in a sample of N = 6460 youth (Nagata, Alsamman, et al., 2023). Heterosexual adolescents walked 676 more steps/day than sexual minority youth. Black adolescents recorded 329 more steps/day than White peers. Fitbit-based monitoring provided objective, repeat measures, revealing nuanced inequities across gender, income, race, and sexuality. Another analysis (Nagata, Sui, et al., 2025) examined gender diversity and physical activity disparities via Fitbit's sensor-based measures (N = 6038). Average steps across the sample were ~9096/day, yet transgender adolescents recorded 1394 fewer steps/day compared with cisgender peers, even after covariate adjustment. Greater gender non-contentedness across multiple domains also predicted lower steps, highlighting the consistency of activity disparities among gender-diverse youth. Remote sensing data captured these sustained patterns objectively over weeks, reducing reliance on recall-based measures. Findings underscore how Fitbit-based sleep and step counts can reveal persistent inequities in behaviors across visible and invisible identity characteristics.

In Harriman et al. (2024), how structural inequities shape objective sleep was assessed (n = 5901). Youth exposed to cumulative disadvantages including housing insecurity, legal challenges, and material hardship recorded significantly shorter nightly sleep duration on Fitbit and elevated parent-reported sleep disturbances compared with peers

experiencing fewer inequities. Distinct inequity classes revealed systemic and interpersonal risks as particularly detrimental to sleep health. Remote sensing data allowed precise detection of these gradients beyond traditional survey measures, underscoring Fitbit's ability to capture the physiological consequences of inequity in adolescence. Another study (Lewis-de Los Angeles, 2024) also assessed links between ACEs and objective sleep and activity (N = 4151 for sleep, N = 6227 for steps). Children exposed to 2 + ACEs slept 6.2 fewer minutes per night and 2.3 fewer minutes of REM sleep, compared with peers with no ACEs. In terms of physical activity, 2 + ACEs were associated with ~417 fewer steps/day, while even a single ACE corresponded to ~323 fewer steps. Fitbit measures highlighted the additive impact of adversity across both domains, providing precise evidence that ACEs undermine health-related behaviors as early as preadolescence.

Fitbit measures have also been investigated as a moderator of health disparity associations. In particular, analyses by Wang and colleagues (2025) tested whether sleep buffered the association between discrimination and later substance-use intentions (N = 1404). Experiences of discrimination, including race/ethnicity, sexual orientation, and weight, predicted higher intentions to use substances. However, adolescents who averaged longer weekday sleep on Fitbit (≥7.5 h) did not show significant increases in intentions, suggesting adequate sleep provided protection. In contrast, peers with shorter objectively measured sleep showed stronger discrimination–substance use links. Regarding physical activity, Al-Shoaibi and colleagues (2024) found strong, graded associations between ACE exposure and reduced physical activity as measured by Fitbit in N = 7046 ABCD participants. Adolescents with four or more ACEs walked 567 fewer steps/day than peers with no ACEs. Specific ACE subtypes also predicted reduced steps, with emotional abuse linked to 719 fewer, physical neglect to 424 fewer, and household mental illness to 317 fewer daily steps. The results demonstrated that different forms of adversity distinctly contributed to reduced movement. Wearable-based tracking allowed the study to map dose–response adversity effects across weeks.

A consistent theme in health equity research is how to reduce biases in order to more accurately assess outcomes and find addressable solutions. Based on the reviewed studies, device-based measures could enhance future research by providing unbiased estimates of activity and sleep across days and intensities. Findings demonstrate Fitbit's value for identifying sleep thresholds relevant to health disparities research. There is also potential for sensing to advance equity-oriented studies on discrimination resilience. Results are robust in many areas (e.g., demographically related sleep gaps). Taken together, device-based results highlight inequities across multiple characteristics and physical health factors, emphasizing the importance of using objective, continuous monitoring. Given sleep and physical activity are modifiable factors, these findings may drive intervention to reduce health disparities.

14. Study quality assessment

Study quality and risk of bias ratings, assessed according to the NIH Quality Assessment Tool for Observational Cohort and Cross-Sectional Studies are provided in full in the Supplementary Material and further summarized in **Appendix A**. Of the 45 reported studies, 21 were rated as "Fair" (46.7%), 18 were rated as "Poor" (40.0%), and 6 as "Good" (13.3%) quality. Frequently observed methodological concerns contributing to "Fair" or "Poor" study ratings included limited statistical control for potential confounding variables, inadequate consideration of missing data or sample selection biases, and psychometric limitations in outcome or exposure variables, most frequently the inclusion of survey measures with limited evidence for reliability or validity. The minority of studies rated as "Good" quality generally included more extensive efforts to address missing data and selection biases, the inclusion of sensitivity analyses, such as replication in external cohorts, which improved evidence for robust, generalizable results, and study designs which reduced susceptibility to between-subjects confounding (e.g., within-subjects analyses). Examining study ratings across (non-exclusive) content areas, we observed the highest proportion of articles rated as "Good" in the "Machine Learning" grouping, (2/4, 50.0%) whereas no "Good" rated studies were identified in either the "Screen Usage" (0/7) or "Physical Health" (0/8) groups. "Poor" ratings were most common in the "Physical Health" (6/8, 75.0%) and "Screen Usage" (4/7, 57.1%) groups.

15. Discussion

This systematic review describes the contribution of ABCD's remote sensor-based wearable device and smartphone use measures to research on adolescent health and development. The review aimed to identify and synthesize all published studies which utilized sensor-based measures from the ABCD study's multi-week remote passive sensing trials. This comprised 45 peer-reviewed empirical research papers published between 2021 and 2025 which utilized either Fitbit-based measures, sensor-based measures of adolescent smartphone use, or both to investigate a broad range of topics related to adolescent health. A significant majority (41/45, 91% of identified articles) exclusively utilized Fitbit data and generally reported modest associations between summary measures of physical activity and sleep with physical and mental health outcomes, psychosocial stressors, and neurocognitive functions. Far fewer articles were identified which utilized ABCD's smartphone sensor measures, as data within the full cohort were only first released in 2024. Preliminary evidence from these studies supports the feasibility and acceptability of data collection, strong psychometric properties, external criterion validity, and relationships with adolescent health and neurocognitive outcomes. Taken together, this literature broadly affirms the value of ABCD's sensor-based measures for providing valid, fine-grained measures of physical activity, sleep, and smartphone use robust to the limitations of self-report.

16. Qualitative synthesis of content domains

Initial efforts assessing the feasibility of Fitbit and smartphone sensor data in ABCD broadly suggest that wearable and sensor based data collection are acceptable to participants, though additional strategies may still be needed to improve recruitment of participants from underrepresented sociodemographic backgrounds (Godino et al., 2020; E. H. Kim et al., 2023; Wing et al., 2022). Lower passive sensing participation rates amongst Black and low-income study participants in particular represent a key interpretive challenge, reducing both the generalizability of findings within these populations as well as rendering findings more susceptible to selection biases.

Validation efforts suggest that ABCD sensor based measures provide reliable and accurate information on participants' smartphone use, sleep and physical activity, though with important caveats that the field lacks feasible alternative gold-standard measures of smartphone use, while comparisons between Fitbit data and research grade-sensors suggest modest errors in most Fitbit measures (Alexander et al., 2024; Godino et al., 2020; Wade et al., 2021; Wing et al., 2022; Zhao et al., 2025).

To date, most substantive studies (e.g., those studies not primarily concerned with measure validation) utilizing the ABCD sensor-based measures have used daily or weekly summary measures of Fitbit physical activity and sleep or smartphone use time logs data. Such studies generally report modest cross-sectional or sparse-longitudinal associations between Fitbit measures and various adolescent health behaviors. These include relationships between physical activity and sleep measures with internalizing, thought disorder and total psychopathology symptoms (Damme et al., 2024; Nelson et al., 2022; Yu et al., 2025; L. Zhang et al., 2024), measures of adolescent obesity and cardiovascular disease risk (De Moraes et al., 2024; Kiss et al., 2024; Nagata et al., 2024; Nagata, Smith, et al., 2023), neurodevelopment, neurocognitive functioning, and exposure to childhood maltreatment and experiences of discrimination and marginalization (Al-shoaibi et al., 2024; Harriman

et al., 2024; Lewis-de Los Angeles, 2024; Nagata, Sui, et al., 2025; Wang et al., 2025). Findings are reported as broadly consistent with the existing self-report literature, suggesting robustness of self-report findings to reporting biases and generalizability in a diverse cross-national adolescent sample.

A number of studies have paired ABCD's remote sensing data with research designs or quantitative methods aimed at improving mechanistic, etiological, and predictive insight into relationships. One paper employed an intensive longitudinal within-subjects design to examine daily and hourly relationships between sensor measures of smartphone use, physical activity, and sleep finding evidence for within-subject, temporally specific effects of excessive smartphone use on daily step counts, physical exercise, sleep duration, and sleep initiation (Alexander et al., 2025). Two studies leveraged genetic methods to gain additional etiological insight into relationships between remote sensing Fitbit measures and clinical outcomes, including identifying novel genetic loci implicated in psychiatric traits and examining pleiotropy and genetic-confounding effects impacting relationships between physical activity, sleep, and psychiatric disorders (Liu et al., 2025; Zhang, Geier, et al., 2025). Others paired sensor measures with neuroimaging data, finding associations between sleep duration and connectivity within canonical functional networks, gray and white-matter structure and integrity, and sensorimotor organization, as well as downstream associations with academic and neurocognitive measures (Michael et al., 2025; Zhang, Schwandt, et al., 2025). Several papers applied machine learning algorithms using dozens or hundreds of Fitbit-derived measures to predict mental and physical health outcomes, including ADHD diagnosis, BMI, and obesity risk, often achieving predictive accuracy above 80% (Kim et al., 2023; Kiss et al., 2024; Liu et al., 2025; Rahman, 2025). Such studies demonstrate that sensor data, while useful on their own, are especially powerful when paired with strong research designs and sophisticated statistical and data-reduction methods, which reduce exogenous confounding, offer additional information into potential genetic or brain-based mechanisms, and maximize signal extraction across dozens of sensor-based measures.

17. Limitations and future directions

Across the ABCD sensor-based literature, a number of limitations were frequently identified. First, Fitbit device measurement properties carry known biases in their measurement of sleep and activity behaviors with step counts and frequently overestimated and sleep and exercise time often underestimated relative to research-grade sensors (Burkart et al., 2021; Farina and Lowry, 2018; Mikkelsen et al., 2020; Tedesco et al., 2019). Additionally, the device model used varied by assessment wave. Fitbit estimates of sleep and physical activity may differ across device types, which may carry different sensors and utilize different estimation algorithms. The authors are unaware of any published research directly comparing the accuracy of different Fitbit charge series devices, though studies comparing research grade actigraphy to Charge HR (Farina and Lowry, 2018), Charge 2 (Tedesco et al., 2019), and Charge 4 (Dong et al., 2022; Waddell et al., 2024) devices find similar biases across models (e.g., greater step counts, reduced energy expenditure, and poor specificity in identifying sleep stages). Nonetheless, the possibility that measurement properties may differ across device generations, operating system updates, or changes to proprietary estimation algorithms remains an important limitation of commercial sensor data.

Second, for EARS, while quantitative duration and timing information is derived by sensors for smartphone use, validation efforts of smartphone sensor measures are limited by a lack of gold-standard alternative measures of smartphone use to compare to ABCD's smartphone sensor measures. Platform heterogeneity has previously limited ABCD's collection of many smartphone use measures to Android users, despite a large majority of U.S. adolescents owning iPhones. Low-income, lower parent education, and non-white participants were underrepresented in sensing trials. Taken together, both the over-sampling of Android users and the under-sampling of some sociodemographic groups carry significant interpretive consequences. In addition to reducing the generalizability of findings within iPhone using youth and underrepresented demographic groups, sampling differences further suggest the presence of selection effects, (e.g., collider biases) which may bias behavioral associations related to the sample selection (Munafò et al., 2018).

Some self-report measures, particularly sleep measures, were misaligned with sensor-based measures, potentially depressing correlations between self-report and sensor-based measures, particularly if youth modified their use or sleep during the sensing trials. Lastly, though data processing is standardized and centrally located, data collection, recruitment, and trial administration occurs across the United States and may vary by site, introducing potential variability which has not been gauged or accounted for in the reviewed studies.

Generally, most of the literature consists of associations between summary measures and behavioral outcomes, with predominantly modest observed effect sizes. Single item summary measures of the sensor data, though straightforward to analyze and robust to the reporting and recall biases of self-report, obscure daily or hourly variation and omit information obtained from other sensor measures. The clinical utility of these small observed relationships, as well as their causal direction, underlying mechanisms, and degree of confounding, are all unclear from these study designs. Longitudinal designs offer some improvements, such as establishing temporal precedence of exposures, though have thus far been rare, as follow-up sensor data were limited prior to the ABCD's 6.0 data release in July 2025.

These study design limitations are reflected in the study's formal quality assessment, where a large majority of studies were rated as "Fair" or "Poor" in quality. Common limitations in this literature included limited statistical or methodological efforts to address potential confounders, insufficient efforts to address missing data or selection effects, and psychometric limitations in non-device-based study measures. In contrast, a small number of highly-rated studies utilized research designs which rendered results more robust to potential confounders, including the use of appropriate statistical controls, extensive efforts to address missing data and selection biases, and robust statistical designs, such as within-person analyses and external replication of results. These high-quality studies may inform best practices for future ABCD passive sensing studies.

A number of future directions are further recommended to further improve validity and representativeness of remote sensing data. First, efforts to further standardize data collection across Android and iOS devices are underway, including working with Apple to allow for parallel collection of daily app usage from an individual's eight most used apps. While smartphone use measures appear largely valid, efforts to collect alternative gold-standard measures (such as via continuous screen recording) in a non-ABCD population would further improve confidence in their validity. Active efforts to engage underrepresented populations in the larger ABCD trial in remote sensing efforts would further improve equity and generalizability of sensor-based results. Efforts to further understand discrepancies between self-report and sensor measures are a particular area of concern. Discrepancies may to some degree represent differences in timing of objective and self-report measures, especially if users are modifying their behavior during the trial. Systematic collection and analysis of concurrent self-report data during sensor-based data collection will help reconcile this difficulty.

Since first being made available in 2021, ABCD remote sensing data have been incorporated in dozens of empirical research projects across a wide variety of adolescent health outcomes. The substantial research output associated with this data source highlights their remarkable generativity and accessibility to adolescent health researchers. Centralized data processing and cleaning and the development of daily and weekly summary measures have further reduced the need for specialized quantitative or data-scientific expertise when working with these data, reducing a key barrier to their use in clinical research. The

wide availability and accessibility of these data to a broad range of clinical researchers are among the clearest strengths of the ABCD remote sensing data. At the same time, such studies are primarily associational studies using single item summary measures of physical activity, sleep, or smartphone use, and have primarily corroborated associations observed in prior self-report literature. Such studies are useful in establishing the robustness of such relationships to reporting and recall effects and shared-method biases, though such findings represent incremental, rather than transformational insights. The small number of studies utilizing temporally granular, multivariate sensor measures, research designs offering stronger statistical control over exogenous confounders, and additional insight into genetic or brain-based mechanisms have yielded promising results. Studies which leverage data science methods, incorporate genetic and neuroimaging tools, and utilize causally-informative analyses, while fewer in number, represent promising future directions for gleaning further insights from these measures.

18. Conclusion

This review contributes an up-to-date account of how the ABCD study's passive sensor measures of adolescent smartphone use, sleep, and physical activity have been used across a wide range of adolescent health research topics. We find that 45 research articles have been published using these data in the last five years. Both Fitbit and smartphone-based data collection are generally reported as feasible and acceptable by research participants. Fitbit data are reasonably consistent with research grade-sensor measures while smartphone use measures are internally consistent, similar to expected usage patterns, and moderately correlated with self-report measures. The sample exhibits important selection biases, most notably the considerable oversampling of Android users (who also provided granular app use measures unavailable for iOS/iPhone users) and the undersampling of black and low SES participants, potentially reducing the generalizability of findings and suggesting some findings may be influenced by selection effects.

Substantively, most studies reported small associations between summary measures of sleep and physical activity with self and parent-reported measures of mental, behavioral, and neurocognitive outcomes in adolescence. Findings are largely incremental and limited by cross-sectional, associational designs. A small number of especially compelling studies and future directions are highlighted. The use of machine learning and other data science methods allow researchers to make maximal use of behavioral information available from granular multivariate sensor measures. Leveraging genetic designs and neuroimaging data offer additional insight into potential biological mechanisms linking mental and behavioral health outcomes to sensor-based measures of physical activity, sleep, and smartphone use. Lastly, causally informative research designs, like within-person difference-in-differences approaches, can, when available, reduce the possibility of third-variable confounding and offer greater insight into the directionality of observed relationships. Overall, the ABCD remote sensing data have proven a highly accessible and generative resource for adolescent health researchers. Future work incorporating other powerful data and methods made possible by this landmark study promise to dramatically advance our understanding of the role of screen-based media use in adolescent physical, mental, and neurodevelopmental health outcomes.

CRediT authorship contribution statement

Jordan D. Alexander: Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Conceptualization. **Sawyer Mills:** Writing – review & editing, Writing – original draft, Formal analysis. **Kelsey Sumpter:** Writing – review & editing, Writing – original draft. **Calen J. Smith:** Writing – review & editing, Writing – original draft, Project administration. **Biplabendu Das:** Writing – review & editing, Writing – original draft, Data curation. **Ruby Fan:** Writing – review & editing, Writing – original draft. **Ethan T. Hunt:** Writing – review & editing, Writing – original draft, Conceptualization. **Sandra A. Brown:** Writing – review & editing, Writing – original draft. **Brenda L. Curtis:** Writing – review & editing, Writing – original draft, Methodology. **Kevin M. Cummins:** Writing – review & editing, Writing – original draft. **Michael Mason:** Writing – review & editing, Writing – original draft. **Michael C. Neale:** Writing – review & editing, Writing – original draft. **Susan F. Tapert:** Writing – review & editing, Writing – original draft, Project administration, Conceptualization. **Fiona C. Baker:** Writing – review & editing, Writing – original draft, Methodology, Conceptualization. **Bernard F. Fuemmeler:** Writing – review & editing, Writing – original draft, Methodology, Conceptualization. **Natasha E. Wade:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Ethics declaration

Written informed consent to take part in the study and to publish the article has been obtained from all participants or their legal representatives. The privacy rights of participants have been observed.

This study was performed in compliance with relevant laws, regulatory frameworks and guidelines where the research took place. This study was approved by the University of California, San Diego. (Approval No. 160091)

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.dcn.2026.101826.

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